LESSON 10
SPEAKING MATHEMATICALLY

Overview:

Students will be introduced to a mathematical way of thinking that can serve them in different situations in this lesson. They will also be introduced to some of the special language of Mathematics such as variable, set, relations, and functions.

Learning Outcomes:

At the end of lesson 10, the students can:

1. translate English sentences to mathematical sentences using variables;
2. write sets using Set-Roster and Set-Builder notations; and
3. differentiate functions from relations.

Materials Needed:

- learning module
- slide presentation
- video lessons

Duration: 6 hours

Learning Content:

10.1 VARIABLE

A **variable** is a symbolic name associated with an object whose associated value may be changed. A **constant** has a fixed value all the time.

Example

1. Are there two numbers with the property that the sum of their squares equals the square of their sum?

Solution

In this sentence you can introduce a variable to replace the potentially ambiguous word "their":

Are there numbers x and y with the property that $x^2 + y^2 = (x + y)^2$?

or: Are there numbers x and y such that $x^2 + y^2 = (x + y)^2$?

or: Do there exist any numbers x and y such that $x^2 + y^2 = (x + y)^2$?

2. Given any real number, its square is nonnegative.

Solution

Given any real number r, r^2 is nonnegative.

or: For any real number r, $r^2 \geq 0$.

or: For all real numbers r, $r^2 \geq 0$.

10.2 IMPORTANT MATHEMATICAL STATEMENTS

We have previously defined a statement as a sentence that is either true or false. Three of the most important kinds of sentences in Mathematics are universal statements, conditional statement, and existential statements.

A **universal statement** says that a certain property is true for all elements in a set.

Example

1. All even numbers are divisible by 2.

Given a property that may or may not be true, an **existential statement** says that there is at least one thing for which the property is true.

Example

2. There is a prime number that is even.

A **conditional statement** says that if one thing is true then some other thing also has to be true.

Example

3. If an animal is a dog, it is a mammal.

Universal Conditional Statements

A **universal conditional statement** is a statement that is both universal and conditional.

Examples

4. For all animal a, if a is a dog, it is mammal.
5. If a is a dog, then a is a mammal.
6. For all dogs a, a is a mammal.

10.3 UNIVERSAL EXISTENTIAL STATEMENTS

A **universal existential statement** is a statement that is universal because its first part says that a certain property is true for all objects of a given type, and it is existential because its second part asserts the existence of something.

Examples

1. Every real number has an additive inverse.
2. All real numbers have additive inverses.
3. For all real numbers r, there is an additive inverse for r.

10.4 SETS

If S is a set, the notation $x \in S$ means that x is an element of S. The notation $x \notin S$ means that x is not an element of S. A set may be specified using the **set-roster notation** by writing all of its elements between braces. For example, $\{\mathbf{1}, \mathbf{2}, \mathbf{3}\}$ denotes the set whose elements are 1, 2, and 3. A variation of the notation is sometimes used to describe a very large set, as when we write $\{\mathbf{1}, \mathbf{2}, \mathbf{3}, \ldots, \mathbf{100}\}$ to refer to the set of all integers from 1 to 100. A similar notation can also describe an infinite set, as when we write $\{\mathbf{1}, \mathbf{2}, \mathbf{3}, \ldots\}$ to refer to the set of all positive integers. (The symbol $\cdots$ is called an **ellipsis** and is read as "and so forth".)

Examples

1. Let $A = \{1, 2, 3\}, B = \{3, 1, 2\}$, and $C = \{1, 1, 2, 3, 3, 3\}$. What are the elements of A, B, and C? How are A, B, and C related?

Solution

$A, B,$ and C have exactly the same three elements: $1, 2,$ and 3. Therefore, A, B, and C are simply different ways to represent the same set.

2. Is $\{0\} = 0$?

Solution

$\{0\} \neq 0$ because $\{0\}$ is a set with one element, namely 0, whereas 0 is just the symbol that represents the number zero.

3. How many elements are in the set $\{1, \{1\}\}$?

Solution

The set $\{1, \{1\}\}$ has two elements: 1 and the set whose only element is 1.

4. For each nonnegative integer n, let $U_n = \{n, -n\}$. Find U_1, U_2, and U_0.

Solution

$U_1 = \{1, -1\}$, $U_2 = \{2, -2\}$, and $U_0 = \{0, -0\} = \{0\}$

Certain sets of numbers are so frequently referred to that they are given special symbolic names. These are summarized in the following table:

Symbol	**Set**
$\mathbb{R}$	set of all real numbers
$\mathbb{Z}$	set of all integers
$\mathbb{Q}$	set of all rational numbers or quotients of integers

The set of real numbers is usually pictured as the set of all points on a line, as shown below.

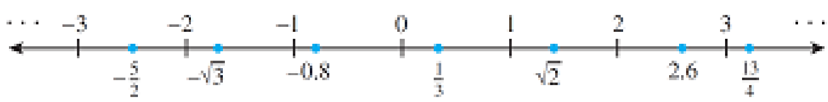


Figure 12.1

The number 0 corresponds to a middle point, called the **origin**. A unit of distance is marked off, and each point to the right of the origin corresponds to a positive real number found by computing its distance from the origin.

Each point to the left of the origin corresponds to a negative real number, which is denoted by computing its distance from the origin and putting a minus sign in front of the resulting number.

The set of real numbers is therefore divided into three parts: the set of positive real numbers, the set of negative real numbers, and the number 0. Note that 0 is neither positive nor negative.

The real number line is called continuous because it is imagined to have no holes.

The set of integers corresponds to a collection of points located at fixed intervals along the real number line.

Thus every integer is a real number, and because the integers are all separated from each other, the set of integers is called discrete. The name discrete mathematics comes from the distinction between continuous and discrete mathematical objects.

Another way to specify a set uses what is called the **set-builder notation**.

Set-Builder Notation

Let S denote a set and let $P(x)$ be a property that elements of S may or may not satisfy. We define a new set to be the set of all elements x such that $P(x)$ is true. We denote this set as follows:

$$\{x \in S \mid P(x)\}$$

the set of all such that

Example

5. Given that $\mathbb{R}$ denotes the set of all real numbers, $\mathbb{Z}$ the set of all integers, and $\mathbb{Z}^+$ the set of all positive integers, describe each of the following sets.
   a. $\{x \in \mathbb{R} | -2 < x < 5\}$
   b. $\{x \in \mathbb{Z} | -2 < x < 5\}$
   c. $\{x \in \mathbb{Z}^+ | -2 < x < 5\}$

Solutions

a. $\{x \in \mathbb{R} | -2 < x < 5\}$ is the open interval of real numbers (strictly) between –2 and 5.

b. $\{x \in \mathbb{Z} | -2 < x < 5\}$ is the set of all integers (strictly) between –2 and 5. It is equal to the set {–1, 0, 1, 2, 3, 4}.

c. Since all the integers in $\mathbb{Z}^+$ are positive, $\{x \in \mathbb{Z}^+ | -2 < x < 5\} = \{1,2,3,4\}$.

Subsets

If A and B are sets, then A is called a **subset** of B, written $A \subseteq B$, if, and only if, every element of A is also an element of B.

Symbolically,

$A \subseteq B$ means that For all elements of x, if $x \in A$ then $x \in B$.

The phrases A is contained in B and B contains A are alternative ways of saying that A is a subset of B.

It follows from the definition of subset that for a set A not to be a subset of a set B means that there is at least one element of A that is not an element of B.

$A \nsubseteq B$ means that There is at least one element of A that is not an element of B

Proper subset

Let A and B be sets, A is a proper subset of B, written as A ⊂ B, if and only if, every element of A is in B but there is at least one element of B that is not in A. Symbolically, A ⊆ B and B ⊈ A.

Example

6. Which of the following are true statements?

a. $2 \in \{1,2,3\}$	c. $2 \subseteq \{1,2,3\}$	e. $\{2\} \subseteq \{\{1\},\{2\}\}$
b. $\{2\} \in \{1,2,3\}$	d. $\{2\} \subseteq \{1,2,3\}$	f. $\{2\} \in \{\{1\},\{2\}\}$

Solution

Only (a), (d), and (f) are true.

For (b) to be true, the set {1, 2, 3} would have to contain the element {2}. But the only elements of {1, 2, 3} are 1, 2, and 3, and 2 is not equal to {2}. Hence (b) is false.

For (c) to be true, the number 2 would have to be a set and every element in the set 2 would have to be an element of {1, 2, 3}. This is not the case, so (c) is false.

For (e) to be true, every element in the set containing only the number 2 would have to be an element of the set whose elements are {1} and {2}. But 2 is not equal to either {1} or {2}, and so (e) is false.

10.5 ORDERED PAIR

Given elements a and b, the symbol (a, b) denotes the **ordered pair** consisting of a and b together with the specification that a is the first element of the pair and b is the second element. Two ordered pairs (a, b) and (c, d) are equal if, and only if, $a = c$ and $b = d$. Symbolically:

$$(a, b) = (c, d) \text{ means that } a = c \text{ and } b = d.$$

Example

7. Is $(1, 2) = (2, 1)$?

Solution

No. By definition of equality of ordered pairs, $(1, 2) = (2,1)$ if, and only if, $1 = 2$ and $2 = 1$. But $1 \neq 2$, and so the ordered pairs are not equal.

10.6 CARTESIAN PRODUCTS

Given two sets A and B, the **Cartesian product** (also called **cross product**)) of A and B, denoted A × B (read "A cross B"), is the set of all ordered pairs (a, b), where a ∈ A and b ∈ B.

$$\mathbf{A} \times \mathbf{B} = \{(\mathrm{a}, \mathrm{b}) \mid \mathrm{a} \in \mathrm{A} \text{ and } \mathrm{b} \in \mathrm{B}\}$$

Example

8. Let $\mathrm{A} = \{1, 2, 3\}$ and $\mathrm{B} = \{\mathrm{u}, \mathrm{v}\}$.
   a. Find A × B
   b. Find B × A
   c. Find B × B
   d. How many elements are in A × B, B × A, and B × B?

9. Let $\mathbb{R}$ denote the set of all real numbers. Describe $\mathbb{R} \times \mathbb{R}$.

Solution

a. $A \times B = \{(1, u), (2, u), (3, u), (1, v), (2, v), (3, v)\}$

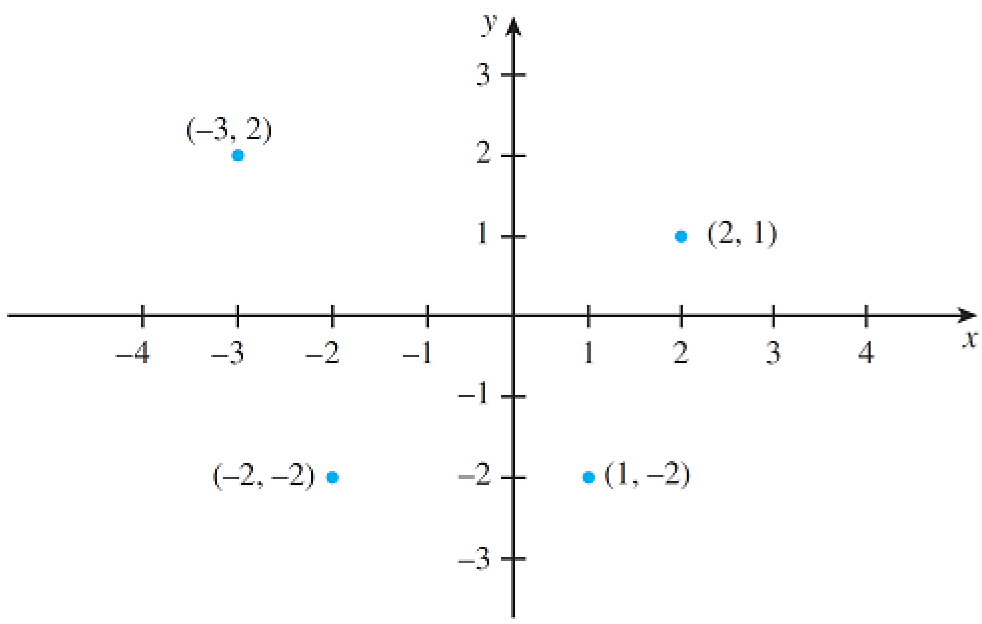
b. $B \times A = \{(u, 1), (u, 2), (u, 3), (v, 1), (v, 2), (v, 3)\}$

c. $B \times B = \{(u, u), (u, v), (v, u), (v, v)\}$

d. $A \times B$ has six elements. Note that this is the number of elements in A times the number of elements in B. B × A has six elements, the number of elements in B times the number of elements in A. $B \times B$ has four elements, the number of elements in B times the number of elements in B.

e. $\mathbb{R} \times \mathbb{R}$ is the set of all ordered pairs (x, y) where both x and y are real numbers. If horizontal and vertical axes are drawn on a plane and a unit length is marked off, then each ordered pair in $\mathbb{R} \times \mathbb{R}$ corresponds to a unique point in the plane, with the first and second elements of the pair indicating, respectively, the horizontal and vertical positions of the point.

The term Cartesian plane is often used to refer to a plane with this coordinate system, as illustrated in Figure 12.1.



A Cartesian Plane

Figure 10.2

10.7 THE LANGUAGE OF RELATIONS AND FUNCTIONS

Relation

Let A and B be sets. A **relation R from A to B** is a subset of $A \times B$. Give an ordered pair (x, y) in A × B, **x is related to y by** $\Re$, written $x\Re y$, if, and only if, (x, y) is in $\Re$. Symbolically,

$$xRy \text{ means } (x, y) \in \Re$$

A – **domain** of $\Re$.

B – **co-domain** of $\Re$.

Example

10. Let A = {1, 2} and B = {1, 2, 3} and define a relation $\Re$ from A to B as follows: Given any $(\mathrm{x},\mathrm{y}) \in \mathrm{A} \times \mathrm{B}$, $(\mathrm{x},\mathrm{y}) \in \Re$ means that $\frac{x-y}{2}$ is an integer.

    a. State explicitly which ordered pairs are in $A \times B$ and which are in $\Re$.

    b. Is $1\Re3$? Is $2\Re3$? Is $2\Re2$?

    c. What are the domain and co-domain of $\Re$?

Solution

a. $A \times B = \{(1,1),(1,2),(1,3),(2,1),(2,2),(2,3)\}$. To determine explicitly the composition of $\Re$, examine each ordered pair in $A \times B$ to see whether its elements satisfy the defining condition for $\Re$.

$(1,1) \in \Re$ because $\frac{1-1}{2} = 0$, which is an integer.

$(1,2) \notin \Re$ because $\frac{1-2}{2} = \frac{-1}{2}$, which is not an integer.

$(1,3) \in \Re$ because $\frac{1-3}{2} = \frac{-2}{2} = -1$, which is an integer.

$(2,1) \notin \Re$ because $\frac{2-1}{2} = \frac{1}{2}$, which is not an integer.

$(2,2) \in \Re$ because $\frac{2-2}{2} = \frac{0}{2} = 0$, which is an integer.

$(2,3) \notin \Re$ because $\frac{2-3}{2} = \frac{-1}{2}$, which is not an integer.

Thus,

$$\Re = \{(1,1),(1,3),(2,2)\}$$

b. Yes, $1R3$ because $(1,3) \in \Re$.

No, $2\not R3$ because $(2,3) \notin \Re$.

Yes, $2R2$ because $(2,2) \in \Re$.

c. The domain of $\Re$ is $\{1,2\}$ and the co-domain is $\{1,2,3\}$.

11. Let $A = \{2,4,5,6\}$ and $B = \{2,3\}$. $\Re$ is a relation from A to B defined by $\Re = \{(x,y)|x \in A, y \in B$ and a is divisible by $b\}$. Find

a. R in the roster form;

b. Domain of $\Re$;

c. Range of $\Re$; and

d. Represent $\Re$ diagrammatically.

Solution

a. $\Re = \{(2,2),(4,2),(6,2),(6,3)\}$

b. Domain of $\Re = \{2, 4, 6\}$

c. Range of $\Re = \{2, 3\}$

d.

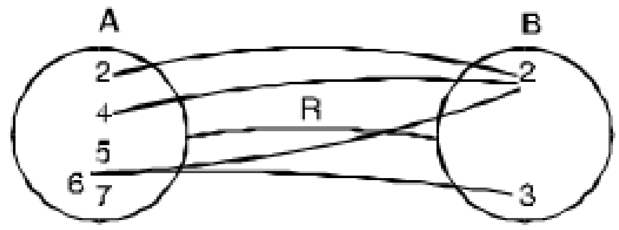


Figure 10.3

Function

Consider the relation

$f = \{(a, 1), (b, 2), (c, 3), (d, 5)\}$

In this relation, we see that each element of Ahas a unique image in B.

This relation f from set A to B where every element of A has a unique image in B is defined as a function from A to B. So we observe that **in a function, no two ordered pairs have the same first elements.**

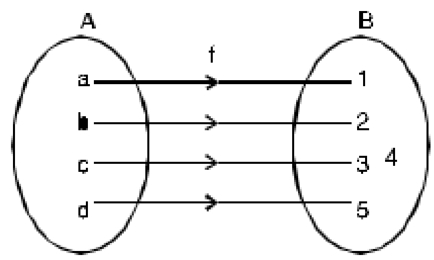


Figure 10.4

We also see that there exists an element 4 in B which does not have its preimage in A. Thus here:

i. the set B will be termed as co-domain and

ii. the set $\{1,2,3,5\}$ is called the range.

From the above we can conclude that **range is a subset of co-domain.**

Symbolically, this function can be written as

$$f : A \rightarrow B \qquad \text{or} \qquad A \xrightarrow{f} B$$

Example

11. Which of the following relations are functions from A to B. Write their domain and range. If it is not a function give reason?

a. $\{(1, -2), (3,7), (4, -6), (8,1)\}$, $A = \{1,3,4,8\}$, $B = \{-2,7, -6,1\}$

b. $\{(1,0), (1, -1), (2,3), (4,10)\}$, $A = \{1,2,4\}$, $B = \{0, -1,3,10\}$

Solution

a. It is a function .

Domain = $\{1,3,4,8\}$, Range = $\{-2,7, -6,1\}$

b. It is not a function. Because its two ordered pairs have the same first elements.

12. State whether each of the following relations represent a function or not.

a.

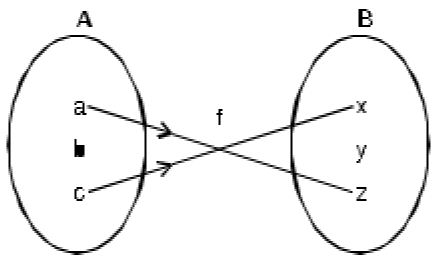


Figure 10.5

b.

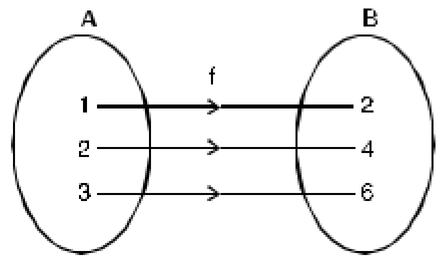


Figure 10.6

Solution

a. f is not a function because the element of A does not have an image in B.

b. f is a function because every element of A has a unique image in B.

13. Which of the following relations from $\mathbb{R} \rightarrow \mathbb{R}$are functions?

a. $y = 3x + 2$ b. $y < x + 3$ c. $y = 2x^2 + 1$

Solution

a. $y = 3x + 2$

Here corresponding to every element $x \in \mathbb{R}$, there exists a unique element $y \in \mathbb{R}$. Hence, it is a function.

b. $y < x + 3$

For any real value of x we get more than one real value of y. Therefore, it is not a function.

c. $y = 2x^2 + 1$

For any real value of x, we will get a unique real value of y. Thus it is a function.

Learning Activity:

1. Write the following sets in set-roster and set-builder notation.
   a. The set of whole number less than 12
   b. The set of odd integers from 3 to 11
   c. The set of positive integers divisible by 4 and less than 58
   d. The set of letters in the word ”MATHEMATICS”
   e. The set of months having 31 days
2. Write in words in how to read each of the following out loud.
   a. $\{x \in \mathbb{R}^+ | \ 0 < x < 1\}$
   b. $\{x \in \mathbb{R} | \ X \leq 0$ or $x \geq 1\}$

c. $\{x \in \mathbb{Z}|$ n is a factor of 6$\}$
d. $\{x \in \mathbb{Z}^+|$ n is a factor of 6$\}$

3. Is $4 = \{4\}$?
4. How many elements are in the set $\{3,4,3,5\}$?
5. Which of the following sets are equal?

   $A = \{0,1,2\}$

   $B = \{x \in \mathbb{R}| -1 \leq x < 3\}$

   $C = \{x \in \mathbb{R}| -1 < x < 3\}$

   $D = \{x \in \mathbb{Z}| -1 < x < 3\}$

   $E = \{x \in \mathbb{Z}^+| -1 < x < 3\}$
6. For each integer n, let $T_n = \{n, n^2\}$. How many elements are in each of T_2, T_{-3}, T_1, and T_0? Justify your answers.
7. Let $A = \{2,3,4\}$ and $B = \{6,8,10\}$ and define a relation R from A to B as follows: For all $(x, y) \in A \times B$,

   $(x, y) \in R$ means that $\frac{y}{x}$ is an integer.

   a. Is $4R6$? Is $4R8$? Is $(3,8) \in R$? Is $(2,10) \in R$?
   b. Write Ras a set of ordered pairs.
   c. Write the domain and co-domain of R.
   d. Draw an arrow diagram for R.

Learning Evaluation:

I. Fill in the blanks using a variable or variables to rewrite the given statement.

1. Is there a real number whose square root is -1?
   a. Is there a real number x such that _______?
   b. Does there exist _______ such that $\sqrt{x} = -1$?
2. Given any real number, there is a real number that is lesser.
   a. Given any real number r, there is _______ s such that s is _______.
   b. For any _______, _______ such that $s < r$.

II. Fill in the blanks to rewrite the given statement.

3. For all real numbers x, if x is an integer, then x is a rational number.
   a. If a real number is an integer, then _______.
   b. For all integers x,, _______.
   c. If x _______, then _______.
   d. All integers x are _______.
4. All real numbers have squares that are not equal to -1.
   a. Every real number has _______.

b. For all real numbers r, there is _______ for r.

c. For all real numbers r, there is a real number s such that _______.

5. There is a positive integer whose square is equal to itself.

a. Some _______ has the property that its _______.

b. There is a real number r such that the square of r is _______.

c. There is a real number r with the property that for every real number s _______.

6. Let A be the set containing all prime numbers less than 30. List down the elements of A.

7. Is $\{2,2\} = \{2,\{2\}\}$?

8. How many elements are in the set $\{a, a, a, a, a\}$?

9. Given that $\mathbb{Z}$ denotes the set of all integers and $\mathbb{N}$ the set of all natural numbers, describe each of the following sets.

a. $\{x \in \mathbb{N} | x \leq 10$ and x is divisible by $3\}$

b. $\{x \in \mathbb{Z} | x$ is a prime and x is divisible by $2\}$

c. $\{x \subseteq \mathbb{Z} | x^2 = 4\}$

10. Let $B = \{2,4,6,8,10\}$, $C = \{4,8,10\}$, and $D = \{x | x$ is even$\}$. Answer the following questions. Give reasons for your answers.

a. Is $D \subseteq B$?

b. Is $C \subseteq D$?

c. Is $C \subseteq B$?

d. Is B a proper subset of D?

11. Let $C = D = \{-3, -2, -1, 1, 2, 3\}$ and define a relation S from C to D as follows:

For all $(x, y) \in C \times D$,

$(x, y) \in S$ means that $\frac{1}{x} - \frac{1}{y}$ is an integer.

a. Is $2S2$? Is $-1S(-1)$? Is $(2,2) \in S$? Is $(2, -2) \in S$?

b. Write Sas a set of ordered pairs.

c. Write the domain and co-domain of S.

d. Draw an arrow diagram for S.

References:

Aufman, R. et. al. (2018). *Mathematics in the Modern World*. Rex Book Store Inc. Sampaloc, Manila, Philippines.

Johnstone, P. (1987). *Mathematics, Logic, Categories, and Sets*. Cambridge University Press.